

# MAYANK SHARMA

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LinkedIn | GitHub | Portfolio | Codeforces

## TECHNICAL SKILLS

**Languages:** JavaScript (ES6+), C++, Python

**Frontend:** React.js, React Router, HTML5, CSS3, Vite

**Backend:** Node.js, Express.js, REST APIs

**Database:** MongoDB, Mongoose

**Security:** JWT, bcrypt

**DevOps & Tools:** Docker, Docker Compose, Git, GitHub, Postman, VS Code

**Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming, Asynchronous JavaScript

**AI:** LLM Integration, OpenRouter API, Prompt Engineering

## PROJECTS

### File Processing System – Scalable CSV Processing Backend

**Tech:** Node.js, Express.js, MongoDB, Mongoose, REST APIs

- Designed a backend system for processing large batches of CSV files and computing **row-wise aggregates across files**, addressing the challenge of coordinating large numbers of input files through a structured batch-processing workflow.
- Built REST APIs using **Node.js and Express.js** for creating processing batches and managing the lifecycle of files and jobs.
- Integrated **MongoDB with Mongoose** to persist batch records, file metadata, job information, processing states, and relationships between uploaded files and processing tasks.
- Designed a **batch and job tracking model** that separates individual file processing from the overall batch, allowing processing state and progress to be tracked at multiple levels.
- Implemented the initial backend structure with modular routes, database models, and request handling to provide a foundation for asynchronous file-processing workflows.

### SmartMoney – Personal Finance Management Platform

**Tech:** React.js, JavaScript, Node.js, Express.js, MongoDB, Mongoose, REST APIs, JWT, bcrypt, Docker, Docker Compose

- Built a full-stack personal finance platform using **React.js, Node.js, Express.js, and MongoDB** for managing expenses, budgets, savings goals, investments, transactions, and financial analytics.
- Designed a modular backend with **REST APIs, Mongoose models, middleware, and protected routes** for handling application data and business operations.
- Implemented authentication using **JWT and bcrypt**, including password hashing, token-based authentication, and access control for protected application routes.
- Developed reusable React components and responsive interfaces using **React Router**, component-based architecture, and structured frontend state and data flows.
- Built interactive financial dashboards and data visualizations to present **spending patterns, budgets, savings progress, investment information, and transaction history**.
- Containerized the frontend, backend, and MongoDB environment using **Docker and Docker Compose**, simplifying local setup and providing consistent development environments.

### Customer Support Deflector – AI-Powered Email Assistant

**Tech:** Node.js, Express.js, JavaScript, OpenRouter API, JWT, REST APIs

- Built an AI-powered customer support assistant using **Node.js, Express.js, and OpenRouter API** to analyze customer emails and generate context-appropriate response suggestions.
- Integrated LLM workflows for **email classification, sentiment analysis, automated reply generation, and tone-based response suggestions**.
- Developed REST APIs with **JWT authentication and protected routes** for managing support data and coordinating AI-assisted workflows.
- Designed a responsive dashboard for reviewing **email analytics, support history, and AI-generated responses**, providing a centralized interface for customer-support workflows.

## ACHIEVEMENTS

- Qualified **AFCAT (Air Force Common Admission Test)** three times.
- Captained **Zonal and District-Level Football Teams**, leading 20+ players in competitive tournaments.

## EDUCATION

**Bachelor of Engineering in Computer Science Engineering (AI & ML)**

Chandigarh University, Mohali, Punjab

2020 – 2024

**CGPA: 7.48/10**